

CO2 - Assessment Tool-1

U. Prabhavathi

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0601	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	U. Prabhavathi	Reg. No.:	192472067
Section / Batch:	CSE-AI	Date:	03-08-26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

90
100

Code of Conduct

I, U. Prabhavathi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: U. Prabhavathi

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	✓	✓		
Application of Concepts	30%	✓	✓		
Problem-Solving Approach	20%	✓	✓		
Accuracy of Solution	20%	✓	✓		
Clarity	10%	✓	✓		
Total Marks (100):		90			

Faculty In-charge	Course Coordinator	Head of Department
Name: <u>Rishu</u>	Name: _____	Name: _____
Signature: <u>[Signature]</u>	Signature: _____	Signature: _____
Date: <u>31/8/24</u>	Date: _____	Date: _____

Q96 Convex Hull - Boundary Points in open Field Mapping

Given Points:-

$(0,0), (2,6), (5,8), (8,5), (7,1), (4,2), (3,4)$

Convex Hull (outer Boundary):

$(0,0) \rightarrow (2,6) \rightarrow (5,8) \rightarrow (8,5) \rightarrow (7,1) \rightarrow (0,0)$

Inner Points (not on Hull):

$(4,2), (3,4)$

Total Comparisons:

7 points $\Rightarrow \frac{7 \times 6}{2} = 21$ Pairs checked

Time complexity:

$O(n^3)$

Result:-

The convex hull is formed by the boundary points:
 $\{(0,0), (2,6), (5,8), (8,5), (7,1)\}$

Q97 Selection Sort - Duplicate Mickle element

Dataset:

18, 27, 18, 42, 9, 93

Passes

Pass: 1

9, 27, 18, 42, 18, 33

Pass: 2

9, 18, ~~18~~, 42, 18, 33

Pass: 3

9, 18, 18, 42, 27, 33

Pass: 4

9, 18, 18, 27, 42, 33

Pass 5:

9, 18, 18, 27, 33, 42

Sorted Array:

9, 18, 18, 27, 33, 42

Comparison: 15

Swaps: 5

Time Complexity:

Best = Average = worst = $O(n^2)$

998 Bubble Sort - Unordered Tail Segment

Dataset:

10, 20, 30, 15, 25, 5

Passes:

Pass 1:

10, 20, 15, 25, 5, 30

Pass 2:

10, 15, 20, 5, 25, 30

Pass 3:

10, 15, 5, 20, 25, 30

Pass 4:

10, 5, 15, 20, 25, 30

Pass 5:

5, 10, 15, 20, 25, 30

Sorted Array:

5, 10, 15, 20, 25, 30

Comparison: 15

swaps: 8

Time complexity:

Best: $O(n)$

Average: $O(n^2)$

Worst: $O(n^2)$

Linear Search - Repeated Value Search

Dataset

13, 26, 39, 26, 52, 65, 78

Target: 26

Search steps

1. $13 \neq 26$

2. $26 = \text{Found}$

First occurrence Position: 2

Comparison: 2

Time Complexity:

• Best: $O(1)$

• Average: $O(n)$

• Worst: $O(n)$

Result: Target found at 2nd Position

Binary Search - Unsuccessful Search Below Minimum Value

Sorted Dataset:

12, 24, 36, 48, 60, 72, 84

Target: 6

Steps

1. Mid = 48 $\rightarrow 6 < 48 \rightarrow$ left half

2. Mid = 24 $\rightarrow 6 < 24 \rightarrow$ left half

3. Mid = 12 $\rightarrow 6 < 12 \rightarrow$ left half

4. Search ends (element not found)

Comparisons: 3

Result: element 6 not found

Time complexity:

• Best: $O(1)$

• Average: $O(\log n)$

• Worst: $O(\log n)$